



MTP Department
SIMHADRI

Note No :1



Ref:SMTTP/MAINT. PLNG./2025-26/MTP/67522

Date:

22/04/2025(18:26:55)

Subject:Approval of revision in LMI on SYSTEM FOR AUDITS OF PTW SYSTEM AT NTPC SIMHADRI

SUB:Approval of revision in LMI on System for Audits of PTW system at NTPC simhadri

LMI/OGN/OPS/SYST/053,REV-01 and APR-2025 is based on the document COS-ISO-00-OGN/OPS/SYST/053 Rev. 1 NOV 2024

The following changes were incorporated in LMI as per revised OGN.

1. Checklist S.no 16 modified as 'Availability of the hard copy of the permits cancelled (operation); Check whether control person signed on cancelled permits'
2. Checklist S.no 20:'Permit validity dates. Check whether permit extension dates are being followed or not' removed
3. Checklist S.no 27: 'Danger tag displayed' added in the text
4. Checklist S.no29:'Whether required special permit was required' added in the text
5. Checklist S.no 45:Modified as 'Check use of N type isolations in permits (Use of N type to be avoided)'
6. Checklist S.no 47:modified as "Compliance of selected persons report for all special permit (working at height permit, permit for floor opening, confined space permit, excavation permit and hot work permit)".
7. Checklist S.no 52:status of pending order for closing to be checked for last 1 year
8. New 'Annex V' SAP T-codes for PTW added
9. Isolation description in the permit to be specific without having any ambiguity. (Random permit copy to be checked). Added at Sl. No. 19 in checklist.

10. Duration column deleted from Annexure – III: Sample format for PTW register.

11. Annexure-II General awareness on PTW system. For Trial run, ROMP is not to be taken. Trial run removed.

Submitted for review and further approval please

Submitted by:

SUSHILA . - 105817 (ENGINEER ,MTP) ,Simhadri
(SUSHILA@NTPC.CO.IN, 8079025234)

On:22/04/2025(18:26:55) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:2

Reviewed & submitted for approval.

Submitted by:

भास्कर वी के कुरीचेटी - उप महाप्रबंधक

Bhaskar V K Kurichety - 008045 (DY. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri

On:22/04/2025(18:37:29) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:3

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,MTP)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:23/04/2025(19:09:53) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:4

Forwarded for approval of revision in LMI

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,MTP)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:23/04/2025(22:02:14) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:5

Reviewed

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:24/04/2025(10:12:31) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:6

LMI reviewed. It is in line with the OGN.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:28/04/2025(11:30:30) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:7

Please check.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:02/05/2025(19:52:47) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:8

In the check list Sl No.10 (Posting of field operators) , Sl No.11 (Utilization of field operators) may be reviewed as field operators are no longer existing.

Sl No.30 (In switchgear, check whether portable earth drains are under the control of operation or not)- The portable earth drains are under the custody of EMD , not in the contro of operation.

Sl N0.40 (Wooden stool for isolation/Normalization in switchgear room)- In Simhadri Fibre stools have been provided in each switchgear for isolation/Normalization.

While preparing LMI,Above points in Checklist are may be changed.

Submitted by:

बी राज शेखर - उप महाप्रबंधक

Batana Raja Sekhar - 008273 (DY. GENERAL MANAGER ,OPERATION)
(BRSEKHAR@NTPC.CO.IN, 9445869087) ,Simhadri

On:06/05/2025(19:28:23) **Ref:**SMTTP/MAINT. PLNG./2025-26/MTP/67522

Note No:9

Updated as suggested by operation

Submitted by:

SUSHILA . - 105817 (ENGINEER ,MTP)
(SUSHILA@NTPC.CO.IN, 8079025234) ,Simhadri

On:08/05/2025(10:14:29) **Ref:**SMTTP/MAINT. PLNG./2025-26/MTP/67522

Note No:10

LMI reviewed and submitted for approval please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:08/05/2025(15:55:31) **Ref:**SMTTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:11

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,FUEL HAND.- OPN)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:15/05/2025(11:27:31) **Ref:**SMTTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:12

Submitted by:

Amulya Kumar Behera - 097433 (ADDL. GENERAL MANAGER ,FUEL TRANSPORTATION)
(AMULYABEHERA@NTPC.CO.IN, 9425534368) ,Simhadri

On:16/05/2025(18:02:43) **Ref:**SMTTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:13

Reviewed and submitted for approval.

Submitted by:

आशीष वर्मा - अपर महाप्रबंधक

Ashish Verma - 008140 (ADDL. GENERAL MANAGER ,FUEL HAND.- MECH MAINT)
(ASHISHVERMA@NTPC.CO.IN, 9650995131) ,Simhadri

On:20/05/2025(16:40:54) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:14

LMI reviewed and forwarded for approval please.

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

On:21/05/2025(11:11:50) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:15

Reviewed and submitted please.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)
(TVRSEKHAR@NTPC.CO.IN, 9650998176) ,Simhadri

On:23/05/2025(18:03:44) **Ref:**SMTTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:16

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:24/05/2025(09:46:56) **Ref:**SMTTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:17

Reviewed,forwarded for approval

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:24/05/2025(11:14:50) **Ref:**SMTTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:18

Submitted by:

अजय गिरि - अपर महाप्रबंधक

Ajay Giri - 007629 (ADDL. GENERAL MANAGER ,FGD)

(AJAYGIRI@NTPC.CO.IN, 7440228332) ,Simhadri

On:24/05/2025(18:25:42) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:19

Reviewed and submitted for necessary approval.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,ELECT MAINT)

(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:24/05/2025(18:53:55) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:20

LMI has been reviewed and found inline with the OGN. It may be processed for further approval.

Submitted by:

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

Suresh Babu Kolla - 007980 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)

(SBKOLLA@NTPC.CO.IN, 9490307556) ,Simhadri

On:26/05/2025(10:42:55) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:21

LMI reviewed and submitted for approval.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MAINTENANCE)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:26/05/2025(16:35:09) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:22

Changes done in Sl No.30 of Annexure-I may please be reviewed.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:26/05/2025(17:22:57) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:23

Reviewed and submitted for approval.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:26/05/2025(17:32:37) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:24

Reviewed and found OK.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:30/05/2025(12:58:09) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:25

LMI is in line with OGN. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:31/05/2025(18:56:08) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:26

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri

On:01/06/2025(10:41:58) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

Note No:27

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

Sameer Sharma - 005041 (EXECUTIVE DIRECTOR ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri

On:02/06/2025(09:30:36) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/67522

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**SYSTEM FOR AUDITS OF PTW SYSTEM
AT
NTPC SIMHADRI**

**LMI/OGN/OPS/SYST/053,
REV-01, APR-2025**

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1.TITLE OF LMI	SYSTEM FOR AUDITS OF PTW SYSTEM AT NTPC SIMHADRI
2.LMI NO.	LMI/OGN/OPS/SYST/053, REV- 01, APR -2025.
3.REFERENCE DOCUMENT	COS-ISO-00-OGN/OPS/SYST/053 Rev. 01, Nov 2024
4.SUPERSEDED DOCUMENT	LMI/ISO-00-OGN/OPS/SYST/053, Rev. 0, JUN 2024
5.PREPARED BY:	Sushila, Engineer (MTP)
6.CHECKED BY:	Jayamol Rachel Varughese, AGM (MTP) Somnath Bhattacharya, AGM (OPN)
7.REVIEWED BY:	Prasanta Kumar Jena, GM(O&M)
8.APPROVED BY:	Sameer Sharma, HOP(SIMHADRI)

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TITLE	SYSTEM FOR AUDITS OF PTW SYSTEM AT NTPC SIMHADRI
LMI NO.	LMI/OGN/OPS/SYST/053, REV-01

1.0 INTRODUCTION

- PTW audit will bring about the deviations in PTW systems & process for corrective action plan depicting responsibility and implementation of the PTW system, to achieve safe compliance with respect to PTW system & creating safe environment for maintenance work.
- In order to realize the above, guidelines & detailed checklist for conducting PTW audits are required for streamlining the PTW system.
- Periodic audit (internal, external & third party) is required to check the completeness with reference to TC documents, statutory requirement, site- specific OEM recommendation etc. and implementation status in terms of deviations for corrective action. This document will be useful tool for conducting the PTW audit.

2.0 SUPERSEDED DOCUMENTS: LMI/ISO-00-OGN/OPS/SYST/053, Rev. 0, JUNE 2024

3.0 DETAILS OF REVISION:

Sl.n o.	Rev. 0	Rev. 01
1.	--	Isolation description in the permit to be specific without having any ambiguity. (Random permit copy to be checked). Added at Sl. No. 19 in checklist
2.	Checklist Sl. No. 20 Permit validity dates. Check whether permit extension dates are being followed or not	Checklist Sl. No 20 deleted as PTW validity period has been removed.
3.	(ZPTWREG T-CODE, select pending for untagging /pending for TECO)	T-code ZPMPLAN_BULK to know the status of PM which stopped auto triggering at Sl. no 21
4	Checklist Sl. No. 27-Check whether module of live permit is locked or not	Check whether module of live permit is locked, or not & Danger Tag is displayed.

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5	Checklist Sl. No. 28 Check whether LWC print out is available with maintenance & operation. To ensure online LWC practice is not being followed.	Checklist Sl. No. 28 modified with "Check whether all types of permits are printed"
6	Checklist Sl. No. 29 modified- Adequacy of the permit taken and work being carried out	Checklist Sl. No. 29 Adequacy of the permit taken, and work being carried out. Whether required special permit was required.
7	Checklist Sl. No. 45 Tagging of permits with "N" to be avoided. Check to be done if permits isolation done with "N" CRITERIA even though equipment is available. If templates are used, N- type would be minimized. If equipment is not available, advise them to get them created through ERP-PM Team. If at all N-type is used, the nomenclature should be specific and unique.(like FL)	Checklist Sl. No. 45 modified as "Check use of N type isolations in permits (Use of N type to be avoided)"
8	Checklist Sl. No. 47 Completeness of checklists in SAP for all special permits (working at height permit, permit for floor opening, confined space permit, excavation permit and hot work permit)	Checklist Sl. No. 47 modified as "Compliance of selected persons report for all special permit (working at height permit, permit for floor opening, confined space permit, excavation permit and hot work permit)".
9	Checklist Sl. No. 52 Pending order for closing (from 01.01.2020 onwards)	Checklist Sl. No. 52, status of pending order for closing to be checked for last 1 year
10	--	Duration column deleted from Annexure – III: Sample format for PTW register.
11	--	SAP T-Codes for PTW added at Annexure-V
12	Annexure-II General Awareness on PTW system 1.1 Types of Permits (ii) ROMP (Restoration of Motive Power) e.g. Trial run	For Trial run, ROMP is not to be taken. Trial run removed. (ii) ROMP (Restoration of Motive Power) e.g. Mill Roller setting, CHP Belt mapping, Hoist checking etc.

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13	Checklist Sl. No. 16 "Availability of the hard copy of the permits cancelled (operation)"	Availability of the hard copy of the permits cancelled (operation); Check whether control person signed on cancelled permits
14		Checklist S.no 60 added "Ensure sufficient no of selected persons appointed for each type of special permit"
15	Annexure-II General awareness on PTW system 1.1 Types of Permits (v) HWP (Hot Work Permit)- e.g. Welding	Point no. (v) modified as "Hot work permit -LWC with Hot work & PTW with Hot Work"-e.g. Welding
16	Annexure-II General awareness on PTW system 1.1 Types of Permits (vi) Confined Space Permit e.g. Boiler drum	Modified as "CSP (Confined Space Permit)-LWC with Confined Space & PTW with Confined Space. e.g. Boiler drum"
17	Annexure-II General awareness on PTW system 1.1 Types of Permits (vii) WHP (Work at Height Permit) e.g. Switchyard	Modified as "WHP (Work at Height Permit) - LWC with Work at Height & PTW with Work at Height e.g. Cooling Tower"
18	Annexure-II General awareness on PTW system 1.1 Types of Permits (viii) PFO (Permit for Floor Opening) e.g. Burner area grill opening	Modified as "PFO (Permit for Floor Opening)- LWC with Floor Opening & PTW with Floor Opening TG floor/Firing floor area checkered plate/grill opening"
19	Annexure-II General awareness on PTW system 1.1 Types of Permits (ix) EWP (Excavation Work Permit) e.g. New cable laying, new building excavation works	Modified as "EWP (Excavation Work Permit) e.g. New cable laying, new building excavation works"

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20	Annexure-II General awareness on PTW system 10. Permit for excavation 10.1 Clearance from various departments including FQA (for verification in original drawings)	Modified as "Clearance from various departments including FES (for verification in original drawings)"
21	Annexure-II General awareness on PTW system PTW with ROMP 3.1. For Trial run of equipment after PTW submitted.	For Trial run of equipment after PTW submitted. Deleted.
22	Annexure-II General awareness on PTW system 6. Hot work PTWs 6.2 Endorsement by CISF personnel	Modified as "6.2 Endorsement by selected person and CISF personnel"

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4.0 PROCESS

4.1. Frequency of PTW audit

PTW audit is to be carried twice in a year, distributed one audit each half of the year. One audit is to be carried out internally & next audit is to be carried out externally. At station level, MTP is responsible for conducting PTW audit. For external, PTW audit team is to be formed by ORT coordinator.

4.2. Formation of Audit Team

ROS /ROS ORT Coordinator is to participate in every external PTW audit. Audit Team for conducting PTW audit shall consist of following members-

4.2.1 Internal PTW audit:

MTP - will be the coordinator for conducting internal PTW audit. Internal audit team shall comprise one member each from Operation, Maintenance, safety (Team member should not be from audited department. For example, if PTW audit is carried out for main plant operation then team member may be from CHP operation or from other stages of the station)

Resp: HOD(MTP)

4.2.2 External PTW audit:

ROS / ROS coordinator is responsible for conducting external PTW audit. The audit team members of operation/ maintenance / MTP / Safety from other stations of same region including ORT coordinator will form the audit team. At least one member from other station other than ORT coordinator is mandatory for external audit team.

Resp: HOD(MTP) / ORT ROS coordinator

4.2.3 Notification of PTW audit observation is to be raised in SAP. In SAP, under SA notification, there is code for PTW audit also. Auditors to enter audit points under

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this code. Similar to Tech audit points can be seen in SAP itself.

4.2.4 Audit report is to be uploaded in respective station website

4.2.5 Audit observations is /are to be reviewed at least once in a month in DPM. Audit exceptions & compliance is also to be discussed during ORT. Audit compliance is to be shared to COS-MMSD group.

4.3. Daily checks in Daily Planning Meeting

4.3.1. Head of Operation shall identify a team of senior executives (should be Senior Authorized Person) to check and identify gaps at random in the PTW issue process. It is to be ensured that at least one person shall be on duty every day.

Resp: HOD (OPN)

4.3.2. One member from the above team shall conduct audit of one PTW daily by selecting PTW randomly from those issued on that particular day.

4.3.3. He shall identify gaps, improvements, procedural correctness, and adequacy of isolations in the PTW, and put up observations during daily planning meeting the next day. This shall ensure better appreciation and understanding of the PTW system

5.0 RECORDS TO BE MAINTAINED

Sl. No.	Record Description	Maintained by	Frequency	Remarks
1	Exceptions of audit point as per checklist	MTP	As per audit schedule	

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6.0 REVIEW: HOD (MTP) will review & take the approval of competent authority(HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

7.0 ANNEXURES:

Annexure – I: PTW audit Checklist

Annexure – II: Brief write-up about PTW system

Annexure – III: Sample format for PTW register

Annexure – IV: Do's & Don'ts of PTW system

Annexure – V: SAP T-codes

Annexure – VI: Abbreviations

TITLE	SYSTEM FOR AUDITS OF PTW SYSTEM AT NTPC SIMHADRI
LMI NO.	LMI/OGN/OPS/SYST/053, REV-01

Annexure – I: PTW Audit Checklist

Dept.		Audit Date			
Auditors					
S.N	Description	Present Status	Action Plan	Responsibility	Target Date
SYSTEM/SAFETY/DOCUMENTATION					
1	Availability of arc suit in sufficient number				
2	Practice of using arc suit during isolation/normalization activity (random checking)				
3	Availability of fuse pullers in sufficient number				
4	Adequate training of operating staff				
5	Availability of non-contact type tester				
6	DC light provision in all switchgears and other places and records of checking				
7	Written procedure for isolation and normalization of switchyard/all switchgears equipment in local language				
8	PTW near misses needs to be reported to COS/ROS - Operation group for sharing				
9	Proper and correct labeling on swgr panel (front and back) and marking of zebra lines on rear side to avoid interchange of panels				

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10	Posting of operators				
11	Utilization of operators				
12	Checking in SAP whether WCD (work clearance documents) template creation is completed or not				
13	Check random PTWs whether WCD template is used or manually created isolation list (WCLE T-code, select plant, and other details & execute. List will be shown. In the current layout add template to display)				
14	List of authorized persons, CP, SAP for PTW system				
15	Availability of the hard copy of the permits issued (to be printed in two copies- Maintenance & Operation copy)				
16	Availability of the hard copy of the permits cancelled (operation); Check whether control person signed on cancelled permits.				
17	Cross permit recording and authentication on all cross permits issued				
	Unclosed Operational WCDs shall form cross permits in the newly issued Operational WCDs. Hence all old permits should have status "Closed" Please ensure Untagging and closure of permits in SAP after work completion				

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18	Use of isolation and normalization slip during issue and return of permit respectively.				
19	Isolation description in the permit to be specific without having any ambiguity. (Random permit copy to be checked)				
20	Availability, appropriateness and completeness of JSA in local language during issue of permit. JSA to be printed along with permit issued to Maintenance department.				
21	Ensure all Orders "TECO" is being done after work completion. Else next PM would not be triggered in Auto. (ZPMPLAN_BULK to know the status of PM which stopped auto triggering)				
22	Check PM Task lists/Plans are available for all Eqpt. Ensure that Plans are active. Random check may be done				
23	Ensure maximum PM/OH Orders are generated from Task list/Plan. Manual creation of orders (especially in OH) is to be avoided and get the Task lists and plans created in SAP for missing eqpt. (IW38 check for orders with blank plan no)				
24	Check whether the signatures of AP, SAP, CP are being followed as per LMI.				

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25	At least one permit for all types to be checked.				
26	Check whether fuse modules of live permit are in safety lockout box/ fuse lockout box or not. Practice of keeping fuses in nearby location is to be avoided.				
27	Check whether module of live permit is locked or not & Danger Tag is displayed				
28	Check whether all types of permits are printed.				
29	Adequacy of the permit taken and work being carried out. Whether required special permit was required				
30	In switchgear check whether portable earth drains are under the control of EMD or not				
31	Check whether list of approved AP from electrical maintenance is available in switchgear for isolation related to HV switching in switchyard.				
32	M3 type notification record				
33	M4 type notifications record				
34	Lock out boxes & key arrangements a) Availability of sufficient quantity (especially required during unit OH) b) Working Condition of lockout boxes c) Ownership of master keys d) Procedures required in case of lost key.				

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35	Locking arrangement a) Locking arrangement of HT switchgear module b) Locking arrangement / Fuse removal of LT switchgear module c) Locking of mechanical equipment such as valves, gates etc. d) Locking & tagging arrangement in DDCMIS.				
36	Confined Space Permit following to be ensured. a) 24 V DC inspection light inside the equipment b) Door opening to be ensured until CSP returned c) No cutting/ welding gas cylinders inside the equipment d) Instrument for measuring oxygen level				
37	Identification & labeling: Equipment description to be painted on all equipment & panels.				
38	Bus coupler to be marked in Red color with safety warning in local language/Hindi/English				
39	Availability of insulation mat near electrical panels.				
40	Wooden/fiber stool for isolation /normalization in switchgear room.				

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TAGGING AND IDENTIFICATION					
41	Tagging of the equipment under permit on the control desk				
42	Tagging of the equipment under permit in the Switchgear				
43	Proper Identification of Switchgear panel backside cover to avoid accidental opening of same when switchgear is not isolated.				
44	Tagging and identification at site as mentioned in JSA				
45	Check use of N type isolations in permits (Use of N type to be avoided)				
DEGREE OF AWARENESS					
46	Degree of awareness about the PTW system and types of PTW to be issued.				
47	Compliance of selected persons report for all special permit (working at height permit, permit for floor opening, confined space permit, excavation permit and hot work permit)				
48	Display of Isolation/ changeover procedure in MCC/Site				
49	Awareness about JSA in work executers at site				
50	Check whether for live permit, permit copy and JSA are available at location. Whether supervisor is aware of hazards in the work being carried out.				

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MISCELLANEOUS					
51	Presence of CISF personnel as specified at the location of Hot Work Permits				
52	Status of pending order for closing (For last 1 year)				
53	Pending notification for closing				
54	Pending permits for untagging				
55	Equipment pending for normalization in SAP (Untagged).				
56	% of emergency job cards raised per month				
ANY OTHER OBSERVATION					
57	Preventing of unauthorized access to switchgears				
58	Review of exceptions of audit points in DPM & ORT as per this document				
59	Check Notification closed without issuing permit & without valid reason may also be mentioned and analysis report attached therein.				
60	Ensure sufficient no of selected persons appointed for each type of special permit				

Signatures of auditors & site representatives.

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Annexure – II: Brief write-up about PTW system

7.1 General awareness on PTW system

1.1 Type of Permits (WCD):

- i. PTW (Permit To work) e.g. Electrical isolation jobs
- ii. ROMP (Restoration of Motive Power) e.g. Mill Roller setting, CHP Belt mapping, Hoist checking etc.
- iii. SFT (Sanction for Test) e.g. Relay testing, Hi-Pot Testing
- iv. LWC (Limited Work Clearance) e.g. Motor greasing, Transmitter online calibration
- v. HWP (Hot Work Permit)- LWC with Hot work & PTW with Hot Work- e.g. Welding
- vi. CSP (Confined Space Permit)-LWC with Confined Space & PTW with Confined Space. e.g. Boiler drum
- vii. WHP (Work at Height Permit) - LWC with Work at Height & PTW with Work at Height e.g. Cooling Tower
- viii. PFO (Permit for Floor Opening)- LWC with Floor Opening & PTW with Floor Opening TG floor/Firing floor area checkered plate/grill opening
- ix. EWP (Excavation Work Permit) - LWC with Excavation Work & PTW with Excavation Work e.g. New cable laying, new building excavation works

1.2 Safety Rules related to PTW (Safety Handbook availability at Control room)

1.3 LMI on PTW system/ Isolation for Plant safety

7.2 Permit to work

7.2.1. For jobs where electrical, mechanical or both isolations are required, and which may affect the operation.

7.2.2. Equipment/ Unit shutdown required

7.2.3. P&I checking

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7.2.4. System isolation required

7.3 PTW with ROMP

7.3.1 Mill, Clinker, Fan & other rotary parts

7.4 SFT

7.4.1 For Relay/CT testing in switchgears

7.4.2 Breaker should be kept in test position

7.5 LWC

7.5.1 Where not all energies are isolated, e.g.: Transmitter online calibration where process side isolation valve is closed, but electrical supply is ON

7.5.2 Isolation may or may not be required

7.6 Hot Work PTWs

7.6.1 Issue in SAP.

7.6.2 Endorsement by selected person and CISF personnel.

7.6.3 Presence of Firefighting personnel at site with firefighting apparatus.

7.7 Confined Space Permit

7.7.1 24 V DC inspection light inside the equipment.

7.7.2 Door opening to be ensured until CSP returned.

7.7.3 No cutting/ welding gas cylinders inside the equipment.

7.7.4 Instrument for measuring oxygen level.

7.7.5 Applicable for Boiler drum internal, Deaerator internal, APH internal & duct internal.

7.7.6 Trained Rescue person availability.

7.8 Work at Height Permit

7.8.1 Endorsement by CFT (cross functional team members)/Scaffolding inspector

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7.8.2 Permit issued after ensuring above

7.9 Permit for Floor Opening

7.9.1 Proper barricading with danger tapes to be ensured

7.9.2 Permit issued after ensuring above

7.10 Permit for excavation

7.10.1 Clearance from various departments including FES (for verification in original drawings)

7.11 Cross permit

7.11.1 For jobs where maintenance work for same equipment is taken up by more than one Planner Group

7.11.2 Recording in PTW register

7.11.3 Recording key nos. properly

7.11.4 Invalid cross permits shouldn't present (due to unclosed old permits) If one permit is already printed & issued and then cross permit is issued later, information to be given to the first party and ensure that hard copy is manually updated with cross permit number.

7.12 Isolations

7.12.1 A complete disconnection which prevents flow, e.g by the removal of a section of pipework, valves of instrument air and oil lines closed and locked, removal of fuses, or electrical switches lock open etc.

7.12.2 A barrier should be inserted to prevent flow, e.g by means of a spade or blank in pipelines. The barrier should be adequate to withstand maximum pressure differential.

7.12.3 A valve should be locked in the closed position with a solid connection between the locking device and the actual pipe closure, or a main electrical switch should be

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locked in the open position

7.12.4 All control, indication and auxiliary power supplies for the valve operating mechanism or electrical switch should be disconnected and such disconnections should be protected against interferences, either by locking off or by appropriately worded warning notices.

7.12.5 All stored energy capable of operating any mechanism within the equipment should be reduced to zero

7.12.6 All test procedures within the limits of isolation should be suspended. Appropriate notices to this effect should be affixed in the area

7.12.7 One permit shall not allow multiple isolations. Specific permit shall be taken for every isolation

7.13 Lock out boxes & key arrangements

7.13.1 Availability of sufficient quantity (especially required during unit OH)

7.13.2 Working Condition of lockout boxes

7.13.3 Ownership of master keys

7.13.4 Procedures required in case of lost key

7.14 Locking arrangement

7.14.1 Locking arrangement of HT switchgear module

7.14.2 Locking arrangement / Fuse removal of LT switchgear module

7.14.3 Locking of mechanical equipment such as valves, gates etc.

7.14.4 Locking & tagging arrangement in DDCMIS.

7.14.5 Locking of manually closed critical valves/dampers.

7.15 Isolation slips

7.15.1 Operator's Name, Signature & Designation

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7.15.2 Recording of Functional location & Equipment details on slip

7.15.3 Procedure to preserve the old slips

7.16 Danger tag

7.16.1 Danger tagging of electrical equipment & LT/ HT switchgear

7.16.2 Danger tag to be tagged properly on mechanical equipment/valves/SWGR

7.16.3 PTW no to be mentioned

7.16.4 Signature and name of person issuing permit

7.17 Identification & labelling:

7.17.1 Equipment description to be painted on all equipment & panels. Check the same if any module is shifted to as pare module. Also, same information to be updated in SAP Functional Location/Eqpt description.

7.18 Equipment normalization

7.18.1 Ensure return of PTW

7.18.2 Simultaneous updating in SAP

7.18.3 Physical inspection of workspace

7.18.4 Normalization slip details

7.19 PTW register:

7.19.1 Recording timely

7.19.2 Recording all the columns properly (eg. Exceptions/ remarks to be filled)

7.19.3 PTW register may be discarded if filled & all the PTW recorded are cancelled.

7.20 Preservation of old PTW hard copies – Hard copies may be preserved for 3 months from the date of cancellation of PTW.

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7.21 SAP procedures

7.21.1 Permit flow in SAP: Created – Prepared – Tag - Tagged - Work Permit Printed/ Rejected – Work Completed – Untag – Untagged – Closed.

7.21.2 Check ZPTWREG T-code in SAP for PTWs pending for TECO/ pending for Untagging.

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Station Name and Unit:[illegible]

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Annexure – IV: Do's & Don'ts of PTW system

In case of BD Notification & Order:

1. BD Notification should be raised on proper functional location & Equipment to identify the correct damage code.
2. While preparing the order against BD Notification, Maintenance user should not change equipment code, Planner group, cost center in order as check is imposed on these fields. If they are required to be changed, before generating the order the notification should be corrected first.
3. Permit should only be applied in order only after ensuring its issue from operation. Once permit is applied in any BD Notification, the total cycle is to be followed before completion of both notification & order.
4. WOCO in the status of Maintenance order is compulsory for closing notification linked to it. Hence ensure "Release for Execution" tab in maintenance order before collecting permit from Control room. "Release for Execution" tab appears in Maintenance order only when operation user completes the tagging cycle (the indicating bulb in all Work approval, Work application, Operation WCD should get converted to green).
5. After taking permit from maintenance order, WCM line in Operations sheet of order should not be deleted by any reason.

In case of PM & OH Maintenance orders:

1. PM & OH of any equipment should be carried out by using proper maintenance type orders like for PM the order type should be PM03 & for OH the order type should be PM08. Any of these jobs should not be carried out against a break down notification.
2. In PM & OH Maintenance orders check is imposed on planner group & cost center. These will be automatically popped up from maintenance plan. These should not be changed manually.

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3. Total permit cycle to be followed if permit is issued by operation. WCM line in operation should not be deleted in case permit is issued by operation. "Release for Execution" tab is to be ensured before collecting permit from Control room. If tab is not found, inform the operation desk engineer for proper tagging in permit.
4. In case the permit is not issued in PM & OH Orders, the work approval tab appearing in maintenance order should be made open (by using cancel assignments) and then WCM line is to be deleted and then the order is to be confirmed.
5. Confirmation using IW41 should only be done after ensuring booking of materials from store's end, booking of services entered in maintenance order, work completion in maintenance order. If any of these are pending, confirmation should not be carried out. Confirmation of the order is the last activity to be performed by maintenance user before technical completion. De-confirmation of the order is not possible by maintenance user as role is withdrawn.
6. After taking permit from maintenance order, WCM line in Operations sheet of order should not be deleted by any reason.
7. ZPMWCA report for monitoring Orphan Permits. There shouldn't be any orphan permits. If any, they should be closed by coordinated efforts of MTP/OPN/MAINTENANCE (if support is required, contact ERP-PM Team).

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Annexure –V: SAP T-Codes for PTW

STANDARD T-CODES	
IP15/IP16	Change/Display Maintenance Plan
IP17/IP18	Change/Display Maintenance Plan Item
IP24	Scheduling overview of Maint Plans
IP19	Scheduling overview & Simulation
IP41/IP01	Create a Maint Plan
IP02	Change a Maint Plan
IA11/IA12	Create/change a PM Tasklist
IA08/IA09	Change/Display List of Tasklists
WCLE	PTW-WCM Process
WCL2	WCD Templates List
WCT1/WCT2/WCT3	Create/Change/Display WCD Template
IW38/IW39	Change/Display Maint Orders
IW21/IW22/IW28	Create/Change/List view of PM Notification
IW31	Create Maint Order (PM03/PM08)
IK01/IK02/IK03	Create/Change/Display Meas Point
IL01/IL02/IL03	Create/Change/Display Functional Location
IE01/IE02/IE03	Create/Change/Display Eqpt
CUSTOM T-CODES	
ZPTWREG	PTW Register
ZPMEQOUT	Eqpt. under permit report
ZPMEQOUT_PDF	
ZPMEQM_REP	Major Eqpt History Report
ZPM_EQNA_STATUS	EQ N/A for > 15 Days
ZOPNORT2	ORT report (PM)
ZEQCSR	Condition Monitoring Status Report
ZPMEQREP	Eqpt dismantling Records

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ZPMHST_REP	Maint Cost Benefit Analysis Report
ZPMEQP_FAIL_HIST	Eqpt Failure History Report
ZPMEQP_REPEAT_FAIL	Eq similar failure repetition report
ZPM_MAT_REP	Eq Material consumption report
ZPMPLAN_BULK	Prev. Mtce Bulk Reports
ZOHPLAN_BULK	OH Mtce Bulk Transactions/Reports
ZWCL2	WCD Templates list with isolations
ZPMSKIP	Skip PM during Unit OH
ZOPI_PROCESS	OPI Dashboard

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Annexure –VI: Abbreviations

CFT: Cross Functional Team

CISF: Central Industrial Security Force

FQA: Field Quality Assurance

LT/HT: Low Tension/High tension

BD: Break Down

PM: Preventive Maintenance

WCM: Work Clearance Management

LMI: Local Management Instruction

OEM: Original Equipment Manufacturer

TC: Technical Compliance

WCD: Work Clearance Document

JSA: Job Safety Analysis

TECO: Technically Complete

MTP: Maintenance Planning

ORT: Operation Review Team

ROS: Regional Operation Services

CHP: Coal Handling Plant

CP: Competent Person

SAP: Senior Authorized Person

AP: Authorized Person

MCC: Motor Control cubicle

DPM: Daily Planning Meeting

PTW: Permit To work

EWP: Excavation Work Permit

ROMP: Restoration of Motive Power

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SFT: Sanction for Test

LWC: Limited Work Clearance

HWP: Hot Work Permit

CSP: Confined Space Permit

WHP: Work at Height Permit

PFO: Permit for floor opening